

Example Problems

Find weight vector $w \in \mathbb{R}^n$ that minimizes the cost function $J(w) = w^T A w + b^T w + \alpha$ where $A \in \mathbb{R}^{n \times n}$ is a symmetric matrix, $b \in \mathbb{R}^n$ is a vector, $\alpha \in \mathbb{R}$.

→ What is the critical point w^* ?

$$J(w) = w^T A w + b^T w + \alpha.$$

$$\begin{aligned} \nabla J(w) \big|_{w^*} = 0 &\Rightarrow 2Aw^* + b = 0 \\ &\Rightarrow w^* = -\frac{1}{2} A^{-1} b \end{aligned}$$

→ Write down the gradient update rule with step size ϵ , starting from $w^{(0)}$.

$$w^{(k+1)} \leftarrow w^{(k)} - \epsilon \underbrace{\nabla f(w^{(k)})}_{\downarrow}$$

$$w^{(k+1)} \leftarrow w^{(k)} - \epsilon (2Aw^{(k)} + b) \rightarrow$$

Define error in an iterate $w^{(k)}$ to be

$$e^{(k)} = w^{(k)} - w^*$$

Write $e^{(k+1)}$ as some $-\int^2 dt e^{(k)}$

$$e^{(k+1)} = w^{(k+1)} - w^*$$

$$\begin{aligned}
& \rightarrow = w^{(k)} - \epsilon (2Aw^{(k)} + b) - w^* \\
& = \underline{e}^{(k)} + w^* - \epsilon (2A \underline{w}^{(k)} + b) - w^* \\
& = e^{(k)} + \cancel{w^*} - \epsilon \{ 2A(e^{(k)} + w^*) + b \} \cancel{w^*} \\
& = (I - 2\epsilon A) e^{(k)} - \epsilon (2Aw^* + b) \\
& = (I - 2\epsilon A) e^{(k)} \quad \left| \nabla J(w) \right|_{w^*} = 2Aw^* + b = 0
\end{aligned}$$

→ What ensures that each successive error $e^{(k+1)}$ is shorter than $e^{(k)}$?

Eigendecomposition

$$A = V \Lambda V^T \quad \left| \quad \Lambda = \begin{bmatrix} \lambda_1 & 0 & 0 & \dots & 0 \\ 0 & \lambda_2 & 0 & \dots & 0 \\ \vdots & & & & \\ & & & & \lambda_n \end{bmatrix} \right.$$

$$e^{(k+1)} = (I - 2\epsilon V \Lambda V^T) e^{(k)}$$

$$= (VV^T - 2\epsilon V\Lambda V^T) e^{(k)}$$

$$= V(I - 2\epsilon\Lambda) V^T e^{(k)}$$

$$I - 2\epsilon\Lambda = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 2\epsilon\lambda_1 & 0 & 0 \\ 0 & 2\epsilon\lambda_2 & 0 \\ 0 & 0 & 2\epsilon\lambda_3 \end{bmatrix}$$

$$= \begin{bmatrix} (1-2\epsilon\lambda_1) & 0 & 0 \\ 0 & 1-2\epsilon\lambda_2 & 0 \\ 0 & 0 & 1-2\epsilon\lambda_3 \end{bmatrix}$$

If each eigenvalue λ_i satisfies $0 < \lambda_i < \frac{1}{\epsilon}$

$$\Rightarrow -1 < \gamma_i < 1$$

$$\gamma_i = 1 - 2\epsilon\lambda_i$$

$V^T e^{(k)} \rightarrow$ same length as $e^{(k)}$

$(I - 2\epsilon\Lambda) V^T e^{(k)} \rightarrow$ is shorter in length than $e^{(k)}$

$\Rightarrow \rho^{(k+1)}$ is also shorter than $\rho^{(k)}$

$$0 < \lambda_i < \frac{1}{\epsilon}$$

Dual of SVM (linear)

Given separable data (x_i, y_i) , $y_i \in \{-1, 1\}$

We seek a separating hyperplane

$$w^T x + b = 0$$

↳ maximize the margin $\rightarrow \min_{w, b} \frac{1}{2} \|w\|^2$

Subject to $y_i (w^T x_i + b) \geq 1$

↳ Primal problem is convex with strong duality.

↳ Introduce Lagrange's multipliers $\alpha_i \geq 0$.

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2} w^T w - \sum_{i=1}^n \alpha_i [y_i (w^T x_i + b) - 1]$$

$$= \frac{1}{2} w^T w - \sum_{i=1}^n \alpha_i y_i w^T x_i - b \sum_{i=1}^n \alpha_i y_i + \sum_{i=1}^n \alpha_i$$

$$\nabla_w d = w - \sum_{i=1}^n \alpha_i y_i x_i = 0$$

$$\Rightarrow w^* = \sum_{i=1}^n \alpha_i y_i x_i$$

$$b\text{-derivative: } \frac{\partial d}{\partial b} = 0 = - \sum_{i=1}^n \alpha_i y_i$$

$$\sum_{i=1}^n \alpha_i y_i = 0$$

$$\alpha_i \geq 0$$

Complementary Slackness Condition

$$\alpha_i [y_i (w^T x_i + b) - 1] = 0$$

$$\Rightarrow \text{If } \alpha_i > 0 \text{ then } y_i (w^T x_i + b) = 1$$

↳ These are the support vectors.

$$f(\alpha) = \inf_{w, b} [\mathcal{L}(w, b, \alpha)]$$

1

n

$$\rightarrow W = \sum_{i=1} \alpha_i y_i x_i$$

$$\|w\|^2 = \sum_{i,j} \alpha_i \alpha_j y_i y_j (x_i^T x_j)$$

$$\mathcal{L} = \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j x_i^T x_j$$

$$- \sum_{i,j} \alpha_i \alpha_j y_i y_j x_i^T x_j + \sum \alpha_i$$

$$= -\frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j x_i^T x_j + \sum \alpha_i$$

Wolfe Dual Problem

$$\max_{\alpha} \sum \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j x_i^T x_j$$

$$\left. \begin{array}{l} \text{Subject to } \alpha_i \geq 0 \\ \sum_{i=1}^n \alpha_i y_i = 0 \end{array} \right\}$$

$i=1$

Gram matrix

$$x_1, x_2, \dots, x_n \in \mathbb{R}^d$$

↳ a matrix with all pairwise inner products, i.e. $G_{ij} = x_i^T x_j$

$$G = \begin{bmatrix} x_1^T x_1 & x_1^T x_2 & x_1^T x_3 \\ x_2^T x_1 & x_2^T x_2 & x_2^T x_3 \\ x_3^T x_1 & x_3^T x_2 & x_3^T x_3 \end{bmatrix}$$

$$X = \begin{bmatrix} x_1^T \\ x_2^T \\ \vdots \\ x_n^T \end{bmatrix} \rightarrow \text{Data matrix} \quad (n \times d)$$

$$G = X X^T$$

↳ Can you prove that G is always PSD?

For any vector $z \in \mathbb{R}^n$,

$$z^T G z = z^T X X^T z$$

$$v = X^T z \in \mathbb{R}^n \Rightarrow v^T = z^T X$$

$$z^T G z = v^T v = \|v\|^2 \geq 0$$

$$\Rightarrow G \succeq 0 \rightarrow \underline{\text{PSD}}$$

$$Q_{ij} = y_i y_j x_i^T x_j, \quad Q \text{ is a Gram matrix}$$

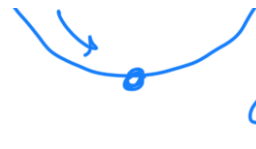
Wolfe Dual Problem:

$$\max_{\alpha} \sum \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j x_i^T x_j$$



$$\left. \begin{array}{l} \max_{\alpha} \mathbf{1}^T \alpha - \frac{1}{2} \alpha^T Q \alpha \\ \text{Subject to } \alpha \geq 0, \quad y^T \alpha = 0 \end{array} \right\}$$





Convex

Concave

Example Problem

$$\min_{x \in \mathbb{R}^2} \frac{1}{2} x^T x = \frac{1}{2} \|x\|^2$$

$$\text{subject to } \alpha^T x = 1, \quad \alpha = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\min_{x_1, x_2} \frac{1}{2} (x_1^2 + x_2^2) \quad \text{subject to } x_1 + x_2 = 1$$

$$x_2 = 1 - x_1 \rightarrow \text{Constraint.}$$

$$f(x) = \frac{1}{2} (x_1^2 + (1 - x_1)^2)$$

$$= \frac{1}{2} (x_1^2 + 1 - 2x_1 + x_1^2)$$

$$= \frac{1}{2} (2x_1^2 - 2x_1 + 1)$$

$$f'(x) = \frac{1}{2} (4x_1 - 2) = 2x_1 - 1 = 0$$

$$\Rightarrow x_1^* = \frac{1}{2}$$

$$\Rightarrow x_2^* = 1 - x_1^* = \frac{1}{2}$$

Primal solⁿ: $x^* = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$

✓ $f(x^*) = \frac{1}{2} \left(\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 \right) = \frac{1}{4} \quad \checkmark = p^*$

$$\begin{aligned} \mathcal{L}(x, \lambda) &= \frac{1}{2} x^T x - \lambda (a^T x - 1) \\ &= \frac{1}{2} (x_1^2 + x_2^2) + \lambda (1 - x_1 - x_2) \end{aligned}$$

Dual function: $g(\lambda) = \inf_x \mathcal{L}(x, \lambda)$

First minimize w.r.t. x .

$\Rightarrow \nabla_x \mathcal{L} = x - \lambda d = 0$

$\Rightarrow x^* = \lambda d = \lambda \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$$\begin{aligned} x^T x &= (\lambda d)^T (\lambda d) = \lambda^2 d^T d \\ &= 2\lambda^2 \end{aligned}$$

$\left| \begin{array}{l} d^T d \\ -1 + 1 \end{array} \right|$

$$\downarrow$$

$$\mathcal{L} = \frac{1}{2} (2\lambda^2) + \lambda (1 - 2\lambda)$$

$$= \lambda^2 + \lambda - 2\lambda^2 = \lambda - \lambda^2$$

$$1 = 2$$

Finally the dual problem

$$\max_{\lambda} (\lambda - \lambda^2)$$

↳ Concave quadratic.

$$\frac{d}{d\lambda} (\lambda - \lambda^2) = -2\lambda + 1 = 0$$

$$\Rightarrow \lambda^* = \frac{1}{2}$$

$$\checkmark g(\lambda^*) = -\frac{1}{4} + \frac{1}{2} = \frac{1}{4} = d^*$$

$$p^* = f(x^*) = g(\lambda^*) = d^*$$

↳ Strong duality holds.

↳ Recover primal solutions as

$$x^* = \lambda^* x = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix}$$

Q. If f be a Convex f^n with Lipschitz Continuous derivatives satisfying

$$\|\nabla f(x) - \nabla f(y)\| \leq L \cdot \|x - y\|$$

$$\forall x, y \in \mathbb{R}^n$$

Prove that :

$$f(y) \leq f(x) + (y-x)^T \nabla f(x) + \frac{L}{2} \|y - x\|^2$$
